

Absolute volume and ball rigidity

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Scope

The absolute Jacobian integral of the radial map yields target-ball volume. A limiting linear contraction has the exact determinant-volume scaling between the two unit balls. Ball inclusion and strict compact-subset volume comparison give ball equality; the separate ball-equality result then gives norm preservation.

5 direct Cards + 28 reused prerequisites = 33 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [The recovered contraction maps one unit ball onto the other](#)
- Key result: [Equality of seminorm balls under a linear bijection preserves the seminorms](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

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Level 0 (11 Cards)

Level 0

A compact proper subset of a seminorm ball has smaller Haar measure

Finds a nonempty open part of the missing set and uses finiteness of the compact set to make the measure comparison strict.

`MathlibAnnex.SeminormBall.measure_lt`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The recovered contraction maps one unit ball onto the other](#)

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Level 0

Equality of seminorm balls under a linear bijection preserves the seminorms

Turns a ball-image equality into one seminorm inequality in each direction.

`MathlibAnnex.SeminormBall.map_eq`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: None in this selected scope

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Level 0

A weighted determinant polynomial with satellites

Adds linear row-replacement contributions to a base determinant.

`MathlibAnnex.Satellite.satellitePolynomial`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The satellite polynomial is linear in the maximal-minor vector](#) · [Absolute maximization forces a near-maximal base](#) · [Each satellite norms its coefficient preimage in absolute value](#)

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Level 0

The family of maximal minors in increasing row order

Collects the determinants of all square submatrices with all columns and an increasing selection of rows.

`MathlibAnnex.Matrix.maximalMinors`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The satellite polynomial is linear in the maximal-minor vector](#) · [Maximal minors scaled by the reference ball volume](#)

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Level 0

A determinant difference bound in the sup operator norm

Controls determinant variation by replacing one matrix row at a time.

`MathlibAnnex.ContinuousLinearMap.abs_det_sub_le_max`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The absolute Jacobian integral of a radial sphere extension](#)

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Level 0

Successive maximum refinement by an ordered list of functionals

Defines a nonempty compact set of survivors after maximizing finitely many functionals in order.

`MathlibAnnex.lexicographicRefine`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A common generator selected from equal convex hulls](#)

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Level 0

Cramer's rule for coordinates of a row

Expresses a row in the basis of the rows of an invertible square matrix through row-replacement determinants.

`MathlibAnnex.Matrix.det_smul_vecMul_nonsingInv_eq_updateRowDet`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Almost-contractive linear recovery with exact volume scaling](#)

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Level 0

Radial extension of an isometry between unit spheres

Extends a bijective sphere isometry to all vectors by retaining each radius and transporting its unit direction.

`MathlibAnnex.Sphere.radialExtension`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The inverse sphere isometry gives the inverse radial map](#) · [The radial extension is 3-Lipschitz](#)

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Level 0

A support face is the convex hull of the maximizing generators

Identifies the points of a convex hull on a supporting level by discarding generators with zero weight.

`MathlibAnnex.convexHull_maxSlice_eq_supportFace`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A common generator selected from equal convex hulls](#)

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Level 0

A seminorm with two-sided bounds against a reference norm

Records a norm through a seminorm, two positive comparison constants and continuity on the original normed space.

`MathlibAnnex.EquivalentSeminorm`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The absolute Jacobian integral of a radial sphere extension](#) · [A strict lower bound on a norm's unit sphere](#) · [Absolute maximization forces a near-maximal base](#) · [Continuous linear maps bounded by a seminorm](#) · [Maximal minors scaled by the reference ball volume](#) · [The recovered contraction maps one unit ball onto the other](#) · [Each satellite norms its coefficient preimage in absolute value](#) · [A nearby norming point gives an almost-norming evaluation](#)

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Level 0

An attained absolute determinant maximum

Selects a maximizing dual frame for a specified basis.

`MathlibAnnex.DeterminantFrame.exists_maximizingFrame`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Absolute maximization forces a near-maximal base](#) · [The set of near-maximal dual frames](#) · [Positivity of the determinant maximum](#)

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Level 1 (12 Cards)

Level 1

The recovered contraction maps one unit ball onto the other

Upgrades ball inclusion to equality by strict compact-subset volume comparison.

`MathlibAnnex.PluckerRecovery.LimitCertificate.image_unitBall_eq`

Immediate Card prerequisites: [A compact proper subset of a seminorm ball has smaller Haar measure](#) · [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: None in this selected scope

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Level 1

Absolute maximization forces a near-maximal base

A dominant determinant term prevents a large deficit in the base frame.

`MathlibAnnex.Satellite.absoluteMaximizer_base_nearMax`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#) · [A weighted determinant polynomial with satellites](#) · [An attained absolute determinant maximum](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#) · [A finite absolute-maximizing family almost norms the unit sphere](#)

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Level 1

Positivity of the determinant maximum

A uniformly scaled dual basis supplies a positive comparison determinant.

`MathlibAnnex.DeterminantFrame.determinantMaximum_pos`

Immediate Card prerequisites: [An attained absolute determinant maximum](#)

Used by in this scope: [A common inverse estimate from row Cramer](#)

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Level 1

Continuous linear maps bounded by a seminorm

Defines the pointwise contraction condition relative to a stored equivalent seminorm.

`MathlibAnnex.EquivalentSeminorm.IsContraction`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [Compactness of the signed generator set](#) · [The symmetric convex body of contraction minors](#)

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Level 1

A strict lower bound on a norm's unit sphere

Defines the strict lower condition on a fixed norm unit sphere.

`MathlibAnnex.FiniteSup.FinMapAlmostIsometric`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#)

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Level 1

The satellite polynomial is linear in the maximal-minor vector

Expresses the configuration polynomial as a linear pairing with maximal minors.

`MathlibAnnex.PluckerSupport.pluckerPairing_satelliteCoefficients_maximalMinors`

Immediate Card prerequisites: [The family of maximal minors in increasing row order](#) · [A weighted determinant polynomial with satellites](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#)

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Level 1

Maximal minors scaled by the reference ball volume

Fixes the volume factor and increasing-row signs.

`MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors`

Immediate Card prerequisites: [The family of maximal minors in increasing row order](#) · [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [Compactness of the signed generator set](#) · [The symmetric convex body of contraction minors](#)

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Level 1

The set of near-maximal dual frames

Records a determinant deficit while keeping all dual rows contractive.

`MathlibAnnex.DeterminantFrame.nearMaxFrames`

Immediate Card prerequisites: [An attained absolute determinant maximum](#)

Used by in this scope: [A common inverse estimate from row Cramer](#) · [Each satellite norms its coefficient preimage in absolute value](#)

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Level 1

The radial extension is 3-Lipschitz

Separates radial variation from the change of unit direction to obtain the global bound with coefficient $1 + 2$.

`MathlibAnnex.Sphere.lipschitzWith_radialExtension`

Immediate Card prerequisites: [Radial extension of an isometry between unit spheres](#)

Used by in this scope: [The absolute Jacobian integral of a radial sphere extension](#)

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Level 1

The inverse sphere isometry gives the inverse radial map

Shows that extending the inverse sphere isometry radially undoes the forward radial extension.

`MathlibAnnex.Sphere.radialExtension_leftInverse`

Immediate Card prerequisites: [Radial extension of an isometry between unit spheres](#)

Used by in this scope: [The absolute Jacobian integral of a radial sphere extension](#)

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Level 1

A nearby norming point gives an almost-norming evaluation

Two triangle inequalities transfer absolute norm attainment to a unit vector.

`MathlibAnnex.Satellite.abs_eval_lower_of_norms_nearby`

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#)

Used by in this scope: [One fixed satellite family almost norms every detected unit vector](#)

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Level 1

A common generator selected from equal convex hulls

Uses a first maximizing slice and a separating sequence to obtain an original common point carrying a prescribed property.

`MathlibAnnex.exists_common_of_convexHull_eq`

Immediate Card prerequisites: [Successive maximum refinement by an ordered list of functionals](#) · [A support face is the convex hull of the maximizing generators](#)

Used by in this scope: [Almost-contractive linear recovery with exact volume scaling](#)

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Level 2 (5 Cards)

Level 2

The absolute Jacobian integral of a radial sphere extension

Identifies the integral of the absolute Jacobian over an open source unit ball with the volume of the closed target unit ball.

`MathlibAnnex.Sphere.radialExtension_integral_abs_det`

Immediate Card prerequisites: [The inverse sphere isometry gives the inverse radial map](#) · [A seminorm with two-sided bounds against a reference norm](#) · [A determinant difference bound in the sup operator norm](#) · [The radial extension is 3-Lipschitz](#)

Used by in this scope: None in this selected scope

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Level 2

A common inverse estimate from row Cramer

Controls all near-maximal inverse frames by an explicit basis-dependent constant.

`MathlibAnnex.DeterminantFrame.inverseBoundConstant_bound`

Immediate Card prerequisites: [The set of near-maximal dual frames](#) · [Positivity of the determinant maximum](#)

Used by in this scope: [A finite absolute-maximizing family almost norms the unit sphere](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

Compactness of the signed generator set

Passes compactness from contractions to their two signed minor images.

`MathlibAnnex.PluckerBody.isCompact_generators`

Immediate Card prerequisites: [Continuous linear maps bounded by a seminorm](#) · [Maximal minors scaled by the reference ball volume](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#)

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Level 2

The symmetric convex body of contraction minors

Takes the real convex hull of both signs of every contraction generator.

`MathlibAnnex.PluckerBody.body`

Immediate Card prerequisites: [Continuous linear maps bounded by a seminorm](#) · [Maximal minors scaled by the reference ball volume](#)

Used by in this scope: [Almost-contractive linear recovery with exact volume scaling](#)

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Level 2

Each satellite norms its coefficient preimage in absolute value

Varying one functional at an absolute maximizer forces endpoint attainment.

MathlibAnnex.Satellite.absoluteMaximizer_satellite_norms_preimage

Immediate Card prerequisites: [A seminorm with two-sided bounds against a reference norm](#) · [A weighted determinant polynomial with satellites](#) · [The set of near-maximal dual frames](#)

Used by in this scope: [One fixed satellite family almost norms every detected unit vector](#)

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Level 3 (1 Card)

Level 3

One fixed satellite family almost norms every detected unit vector

Uses a detecting coefficient and its matching satellite without replacing the family.

MathlibAnnex.Satellite.absoluteMaximizer_satellites_almost_norm

Immediate Card prerequisites: [Each satellite norms its coefficient preimage in absolute value](#) · [A nearby norming point gives an almost-norming evaluation](#)

Used by in this scope: [The chosen support slice consists of almost-isometric generators](#) · [A finite absolute-maximizing family almost norms the unit sphere](#)

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Level 4 (2 Cards)

Level 4

A finite absolute-maximizing family almost norms the unit sphere

Chooses the inverse bound, net and weight before fixing one satellite family.

`MathlibAnnex.Satellite.exists_goodSatellitePackage`

Immediate Card prerequisites: [A common inverse estimate from row Cramer](#) · [Absolute maximization forces a near-maximal base](#) · [One fixed satellite family almost norms every detected unit vector](#)

Used by in this scope: [Almost-contractive linear recovery with exact volume scaling](#)

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Level 4

The chosen support slice consists of almost-isometric generators

Obtains one almost-isometric map from each point of the specified raw maximum slice.

`MathlibAnnex.FiniteRecovery.targetRawSupportMaximizer_good`

Immediate Card prerequisites: [The satellite polynomial is linear in the maximal-minor vector](#) · [Compactness of the signed generator set](#) · [A strict lower bound on a norm's unit sphere](#) · [Absolute maximization forces a near-maximal base](#) · [One fixed satellite family almost norms every detected unit vector](#)

Used by in this scope: [Almost-contractive linear recovery with exact volume scaling](#)

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Level 5 (1 Card)

Level 5

Almost-contractive linear recovery with exact volume scaling

Recovers one bijective linear factor with a norm estimate and exact volume scaling.

`MathlibAnnex.PluckerRecovery.nonempty_linearCertificate_of_pluckerBodies_eq`

Immediate Card prerequisites: [The chosen support slice consists of almost-isometric generators](#) · [The symmetric convex body of contraction minors](#) · [Cramer's rule for coordinates of a row](#) · [A finite absolute-maximizing family almost norms the unit sphere](#) · [A common generator selected from equal convex hulls](#)

Used by in this scope: [A volume-normalized linear contraction obtained by a limit](#)

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Level 6 (1 Card)

Level 6

A volume-normalized linear contraction obtained by a limit

Takes a compact operator limit preserving the exact volume equation.

`MathlibAnnex.PluckerRecovery.nonempty_limitCertificate_of_pluckerBodies_eq`

Immediate Card prerequisites: [Almost-contractive linear recovery with exact volume scaling](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)